

WT Comparisons

Compare Infrastructure-based Network and Adhoc Network.

Factor	Infrastructure-based Network	Adhoc Network
Technology	Access points, switches	Peer-to-peer wireless
Structure	Centralized, fixed	Decentralized, dynamic
Working	Via base station	Direct device communication
Data Rate	Generally higher	Generally lower
Feature Introduced	Central management	Self-configuring nodes
Security	Stronger authentication	Vulnerable to attacks
Interference	Managed centrally	Higher due to hops
Complexity	Lower for nodes	Higher routing complexity
Limitation	Infrastructure dependent	Scalability issues
Applications	Enterprise Wi-Fi	Military, disaster relief

Compare Frequency Hopping Spread Spectrum (FHSS) and Direct Sequence Spread Spectrum (DSSS).

Factor	FHSS	DSSS
Full form	Frequency Hopping Spread Spectrum	Direct Sequence Spread Spectrum
Technology	Carrier frequency hopping	Pseudo-noise coding
Structure	Narrowband, frequency switching	Wideband, continuous signal
Working	Changes frequencies rapidly	Multiplies data with chipping code
Data Rate	Lower (per channel)	Higher
Feature Introduced	Frequency agility	Code division
Security	Good against eavesdropping	Better anti-jamming
Interference	Avoids narrowband interference	Spreads interference energy
Complexity	Moderate (synchronization needed)	Higher (code generation)

Factor	FHSS	DSSS
Limitation	Low throughput	Near-far problem
Applications	Bluetooth, cordless phones	GPS, CDMA, Wi-Fi

Compare FDMA, TDMA and CDMA.

Factor	FDMA	TDMA	CDMA
Full form	Frequency Division Multiple Access	Time Division Multiple Access	Code Division Multiple Access
Technology	Separate frequency bands	Separate time slots	Unique code sequences
Structure	Continuous transmission	Pulsed transmission	Simultaneous transmission
Working	Each call on different freq	Each call in time slice	All on same freq/time
Data Rate	Fixed per channel	Varies per slot	Variable, soft capacity
Feature Introduced	Frequency separation	Time sharing	Code separation
Security	Low (easy to tap)	Moderate	High (code unknown)
Interference	Adjacent channel	No within system	All users interfere
Complexity	Low (simple filters)	Moderate (sync needed)	High (power control)
Limitation	Inefficient use of spectrum	Needs guard times	Near-far problem
Applications	Analog cellular (1G)	GSM (2G)	3G, GPS

Differentiate between GSM and UMTS.

Factor	GSM	UMTS
Full form	Global System for Mobile	Universal Mobile Telecom System
Technology	TDMA + FDMA	W-CDMA
Structure	Circuit-switched	Circuit + Packet
Working	Time/freq slots	Code division
Data Rate	9.6 – 171 kbps	384 kbps – 2 Mbps

Factor	GSM	UMTS
Feature Introduced	SIM card, roaming	High-speed data
Security	Moderate (A5/1 cipher)	Stronger (A5/3)
Interference	Lower (guard bands)	Higher (self-interference)
Complexity	Lower	Higher
Limitation	Low data rate	Needs more bandwidth
Applications	Voice, SMS, GPRS	Video calls, mobile web

Compare 1G to 5G mobile standards.

Factor	1G	2G	3G	4G	5G
Full form	First Generation	Second Generation	Third Generation	Fourth Generation	Fifth Generation
Technology	Analog (AMPS)	Digital (GSM, CDMA)	W-CDMA, UMTS	LTE, WiMAX	NR (mmWave, sub-6)
Structure	Analog cellular	TDMA/FDMA	Packet-switched	All-IP	Cloud-native
Working	Frequency division	Time/freq slots	Code + packet	OFDMA	OFDMA + beamforming
Data Rate	2.4 kbps	64–171 kbps	2 Mbps – 42 Mbps	100 Mbps – 1 Gbps	1–20 Gbps
Feature Introduced	Mobile voice	SMS, digital encryption	Mobile internet	Video streaming	URLLC, mMTC
Security	Very weak	Moderate	Strong	Stronger	Enhanced privacy
Interference	High	Medium	Medium	Low	Very low
Complexity	Low	Medium	High	Very high	Extremely high
Limitation	No data	Low speed	High latency	Complex MIMO	Limited coverage
Applications	Voice calls	Voice + SMS	Web browsing	HD video	IoT, autonomous cars

Compare IEEE 802.11 and IEEE 802.16.

Factor	IEEE 802.11	IEEE 802.16
Full form	Wireless LAN standard	Wireless MAN standard
Technology	CSMA/CA	OFDM, QoS scheduling
Structure	Infrastructure or adhoc	Base station + subscribers
Working	Contention-based access	Scheduled access
Data Rate	11 Mbps – 9.6 Gbps	Up to 1 Gbps (fixed)
Feature Introduced	Wi-Fi	WiMAX
Security	WEP, WPA, WPA3	AES, PKM
Interference	High (unlicensed band)	Lower (licensed possible)
Complexity	Low to moderate	Higher
Limitation	Short range (~100 m)	Higher cost, less mobility
Applications	Home, office WLAN	Backhaul, rural broadband

Compare VANET and MANET.

Factor	VANET	MANET
Full form	Vehicular Adhoc Network	Mobile Adhoc Network
Technology	Dedicated Short Range Comm (DSRC)	Bluetooth, Wi-Fi direct
Structure	Highly dynamic, road-based	Random mobility
Working	Vehicle-to-vehicle (V2V)	Peer-to-peer
Data Rate	3–27 Mbps (DSRC)	1–100 Mbps
Feature Introduced	Roadside units (RSUs)	Fully self-organizing
Security	Critical (safety)	Moderate
Interference	High (dense traffic)	Variable
Complexity	Very high (fast moving)	Moderate
Limitation	Regular disconnections	Limited battery/range
Applications	Collision warning, tolling	Disaster recovery, battlefields